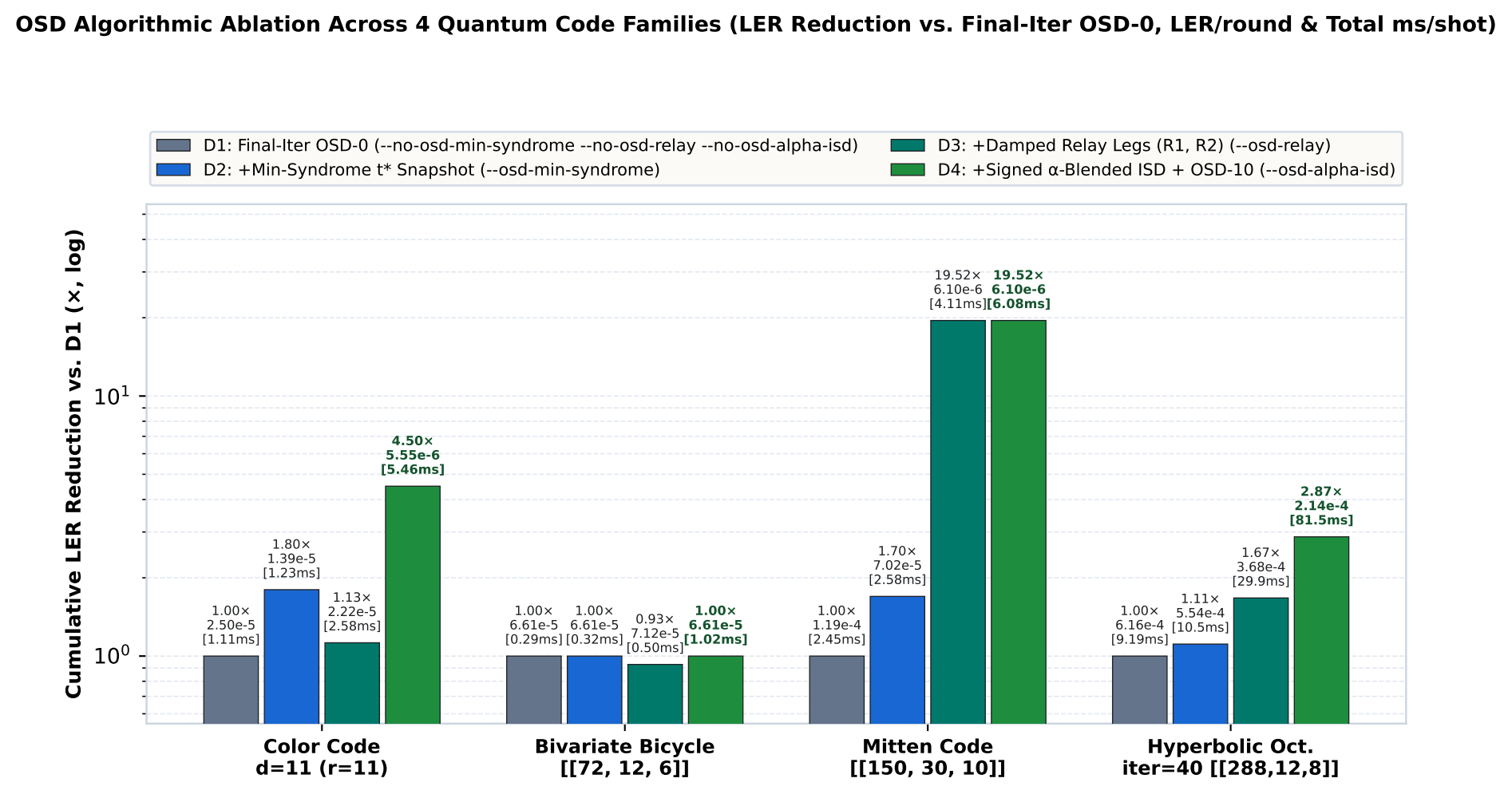
OSD Algorithmic Ablation Across 4 Quantum Code Families (LER Reduction vs. Final-Iter OSD-0, LER/round & Total ms/shot)
D1: Final-Iter OSD-0 (--no-osd-min-syndrome --no-osd-relay --no-osd-alpha-isd)
D2: +Min-Syndrome t* Snapshot (--osd-min-syndrome)
D3: +Damped Relay Legs (R1, R2) (--osd-relay)
D4: +Signed α-Blended ISD + OSD-10 (--osd-alpha-isd)
Cumulative LER Reduction vs. D1 (×, log)
10^1
10^0
1.00× 2.50e-5 [1.11ms]
1.80× 1.39e-5 [1.23ms]
1.13× 2.22e-5 [2.58ms]
4.50× 5.55e-6 [5.46ms]
1.00× 6.61e-5 [0.29ms]
1.00× 6.61e-5 [0.32ms]
0.93× 7.12e-5 [0.50ms]
1.00× 6.61e-5 [1.02ms]
1.00× 1.19e-4 [2.45ms]
1.70× 7.02e-5 [2.58ms]
19.52× 6.10e-6 [4.11ms]
19.52× 6.10e-6 [6.08ms]
1.00× 6.16e-4 [9.19ms]
1.11× 5.54e-4 [10.5ms]
1.67× 3.68e-4 [29.9ms]
2.87× 2.14e-4 [81.5ms]
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]